

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD 402 103
End Semester Examination – 2022

Branch: B. Tech. (Common to all Branches)

Semester: I

Subject with Subject Code: Engineering Mathematics – I (BTBS 101)

Marks: 60

Date:

Time: 3 Hrs.

Instructions to the Students

1. Illustrate your answers with neat sketches, diagrams, etc., wherever necessary.
2. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Q. 1 Solve any THREE:

- (a) Find the rank of a matrix A by reducing it to normal form, where

$$A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}. \quad [4 \text{ Marks}]$$

- (b) Determine the consistency of the set of equations

$$x - 2y + z = -5; \quad x + 5y - 7z = 2; \quad 3x + y - 5z = 1. \quad [4 \text{ Marks}]$$

- (c) Solve the equations:

$$x + 2y + 4z + w = 0; \quad 2x + 4y + 8z + 2w = 0; \quad 3x + 6y + 2z = 0. \quad [4 \text{ Marks}]$$

- (d) Find the eigen values and corresponding eigen vectors of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}. \quad [4 \text{ Marks}]$$

Q. 2 Solve any THREE:

- (a) If $z(x + y) = x^2 + y^2$, show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$. [4 Marks]

- (b) If $v = \log(x^2 + y^2 + z^2)$, prove that $(x^2 + y^2 + z^2) \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right) = 2$. [4 Marks]

- (c) If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$. [4 Marks]

- (d) If $z = f(x, y)$, where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$, then show that

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}. \quad [4 \text{ Marks}]$$

Q. 3 Solve any THREE:

- (a) If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$, show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$. [4 Marks]
- (b) Expand $f(x, y) = e^{xy}$ at $(1, 1)$. [4 Marks]
- (c) Expand $f(x, y) = e^x \sin y$ in the powers of x and y as far as the terms of third degree. [4 Marks]
- (d) Find the maximum value of $x^m y^n z^p$ when $x + y + z = a$. [4 Marks]

Q. 4 Solve any THREE

- (a) Evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$. [4 Marks]
- (b) Evaluate $\int_0^{\frac{\pi}{2}} \sin^8 x \cos^7 x \, dx$. [4 Marks]
- (c) Trace the curve $y^2 = x^3$ (Semi-cubical Parabola). [4 Marks]
- (d) Trace the curve $= a \cos t + \frac{a}{2} \log_e \tan^2 \left(\frac{t}{2} \right)$; $y = a \sin t$ (Tractrix). [4 Marks]

Q. 5 Solve any THREE

- (a) Evaluate $\int_1^a \int_1^b \frac{dydx}{xy}$. [4 Marks]
- (b) Evaluate $\int_0^8 \int_{\sqrt[3]{y}}^2 \sqrt{(1+x^4)} \, dx dy$. [4 Marks]
- (c) Evaluate $\int_0^1 \int_0^y \int_0^{xy} xy^2 z^3 \, dx dy dz$. [4 Marks]
- (d) Find the area bounded by $y^2 = 4x$ and $2x - 3y = -4$. [4 Marks]
